

Bright Coffee Shop Sales Analysis

Methodology

1. Introduction

This project was conducted for the analysis of the Bright Coffee Shop's transactional sales data and provides business insights that support strategic decision-making. The primary objective was to identify revenue drivers, evaluate product performance, understand customer purchasing patterns, and recommend strategies to improve sales and operational efficiency.

2. Data Collection

The project used transactional sales data provided in Microsoft Excel format. The dataset contained information such as:

- Transaction ID
- Transaction Date
- Transaction Time
- Store Location
- Product Category
- Product Type
- Product Detail
- Transaction Quantity
- Unit Price

The Excel file was converted into CSV format to enable processing within Databricks.

3. Data Processing

The dataset was uploaded into Databricks where an ETL (Extract, Transform, Load) process was performed.

Extract

- Imported the CSV dataset into Databricks.
- Verified that all columns were correctly imported.

Transform

Several data cleaning and transformation steps were performed:

- Converted unit_price values from text to numeric format.
- Replaced commas used as decimal separators with decimal points.
- Created a new calculated column called total amount using:

$\text{total_amount} = \text{unit_price} \times \text{transaction_qty}$

- Created a transaction_time_bucket column using 2-hour intervals to simplify time-based analysis.

Examples:

- 06:00–08:00
- 08:00–10:00
- 10:00–12:00
- 12:00–14:00

Load

The cleaned and transformed dataset was stored in Databricks as a processed table ready for SQL analysis.

4. Data Analysis

SQL queries were used to answer the project's business questions.

The analysis included:

- Total Revenue
- Revenue by Product Category
- Revenue by Product Type
- Revenue by Store Location
- Revenue by Time Bucket
- Quantity Sold by Product Category
- Quantity Sold by Product Type
- Identification of Best-Selling Products
- Identification of Underperforming Products

5. Data Visualization

The processed data was exported into Microsoft Excel where Pivot Tables and interactive charts were developed.

Visualizations included:

- Revenue by Product Category
- Revenue by Product Type
- Peak Sales Time Intervals
- Quantity Sold by Product Category
- Store Performance

These visualizations provided a clear understanding of sales trends and business performance.

6. Business Insight Development

The findings were interpreted to identify:

- Products generating the highest revenue.
- Products requiring promotional support.
- Peak customer purchasing periods.
- Opportunities to improve inventory management.
- Sales patterns across store locations.

7. Outcome

The project successfully transformed raw transactional data into meaningful business insights that can assist Bright Coffee Shop in improving operational efficiency, increasing revenue, and making informed business decisions through data-driven strategies.